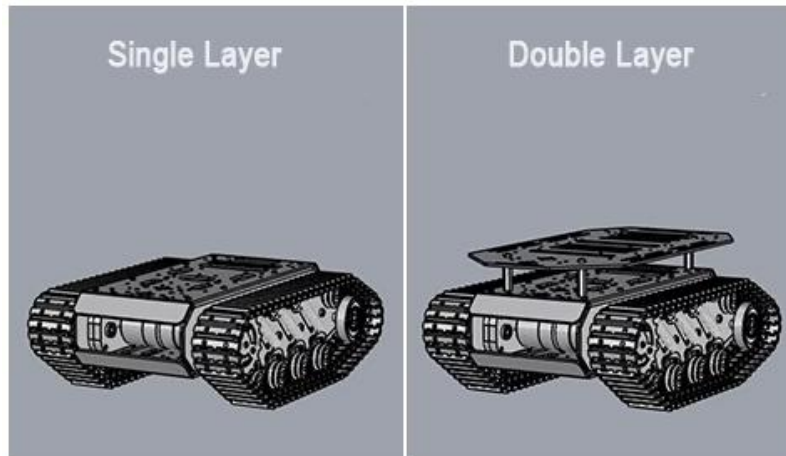


Suspension Track Chassis Introduction

1. Introduction



The suspended track chassis is specifically designed for all-terrain intelligent cars. This chassis features:

Features:

1. Excellent shock absorption performance: Distinct from the standard tracked chassis, it is equipped with 8 channels of highly elastic carbon steel springs and durable miniature bearings.
2. Good rigidity, not easily deformed: It possesses a strong anti-skid ability and can travel stably in various terrain conditions, significantly improving off-road performance.
3. Easy installation: The chassis is made of all-metal material with a structure built using multiple plates. It includes numerous reserved development and expansion mounting holes, making it easy to install various ROS (Robot Operating System) accessories such as Lidar, depth cameras, and microphone array. This greatly facilitates development and expansion work.
4. Expandability: It enables the car to make some autonomous movements

by capturing environmental information around it, providing more advanced functionality.

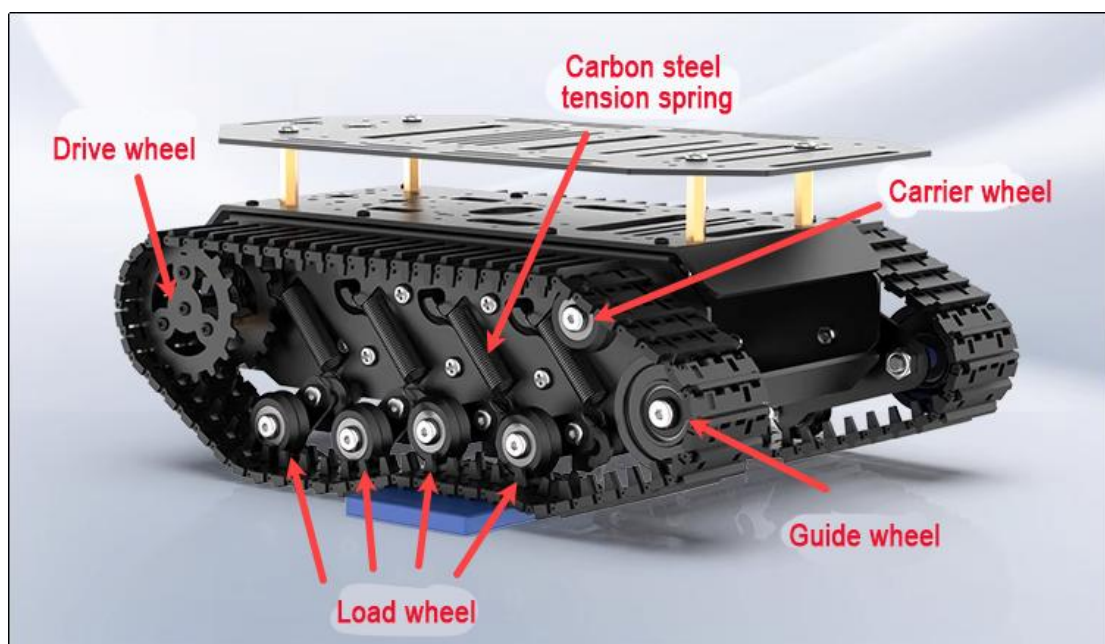
In this course, we learn about the assembly of accessories needed for assembling the car and their connection methods from the hardware structure. From a programming perspective, we can learn how to control the car chassis motion through different control boards, understanding the connection between software and hardware.

2. Introduction to Track Chassis Hardware

❖ To access detailed specifications for the tracked chassis, please navigate to the 'Specification & Drawing\ Technical Specification' section.

2.1 Hardware Structure

Typically, a track chassis is propelled by an drive wheel and utilizes a flexible track loop that encompasses drive wheels, load wheels, guide wheels, and carrier wheels.



- **Drive Wheel**

As a crucial part of the tracked propulsion system, the drive wheel is responsible for transmitting the power from the transmission device to the track. In this product, the drive wheel is located at the front of the vehicle, playing a vital role in effectively transferring power to the track.

- **Load Wheel**

The main function of the load wheel is to support the weight of the vehicle body and maintain the correct position of the track. It helps distribute track pressure on the ground and enhances driving stability.

- **Guide Wheel**

The position of the guide wheel is installed according to the position of the drive wheel, usually in front of the vehicle body. Mainly used to guide the track to wind correctly to prevent deviation and derailment.

- **Carrier Wheel**

The role of the carrier wheel is to uphold the track and prevent it from excessive sagging or side-slipping. This helps minimize vibration and track pulsation during movement.

- **Continuous Track**

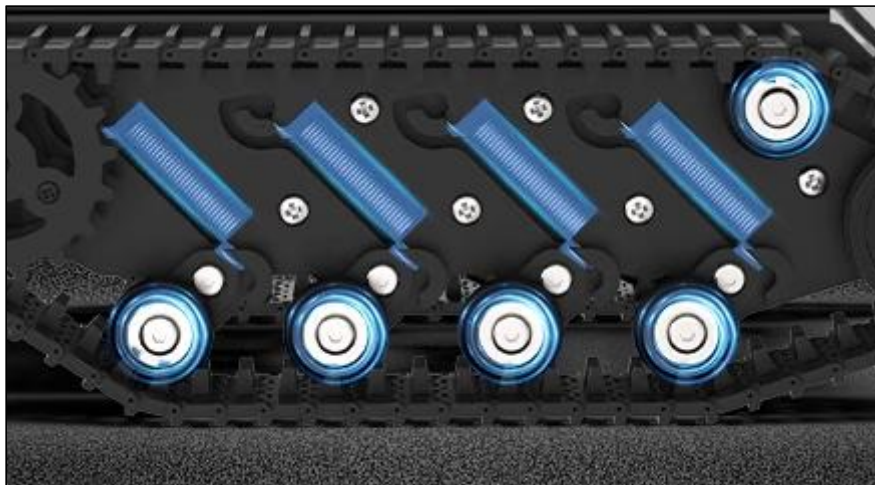
The track is composed of multiple track shoes connected by track pins to create a continuous chain link structure. This design ensures the flexibility and

durability of the tracks.

- **Chassis**

A track chassis moves by means of tracks on its base. During travel, the motion of the track chassis is akin to walking along a pre-laid path. Power is transmitted through the drive wheel, propelling the track to move continuously. The track is unloaded from the guide wheel, laid on the ground, and advances as it rolls under the pressure of the load wheel. Simultaneously, the portion of the track that has passed the load wheel is re-rolled by the drive wheel.

2.2 Special Design



Compared to traditional track chassis, this suspension tracked vehicle features unique design elements. The key innovation involves additional suspension fixtures on the load wheels, equipped with high-elastic carbon steel tension springs, providing the load wheels with a certain range of movement. This design not only ensures balanced pressure distribution and stable power transmission but also significantly enhances the shock-absorbing capacity of the car chassis by pairing it with high-durability micro-bearings. The vehicle

effectively absorbs vibrations from uneven terrain, reducing impact on the chassis and load. Additionally, this design enhances the vehicle's adaptability to various terrains, enabling it to easily traverse complex landscapes and ensure smooth driving. For detailed technical parameters and drawings of this tracked vehicle, please refer to the '3. Specification & Drawing' section.

3. Development Tutorials Description

The development tutorials are edited based on different controllers, including Arduino Uno, STM32 controller, and Raspberry Pi 4B. You can select the appropriate course based on your specific controller.